

# The Collatz Conjecture and Non-Trivial Cycles; Conference Abstract

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## 1 Introduction

The Collatz Conjecture, also known as the  $3x+1$  problem, is one of mathematics' most infamous unsolved problems. Given infinite repetition of the following rules:  $n \rightarrow 3n + 1$  if  $n$  is odd, and  $n \rightarrow n/2$  if  $n$  is even, the Collatz Conjecture states that all natural numbers reach one, entering the trivial cycle  $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$ . Given the simple arithmetic rules being applied, circumstantial evidence, etc., mathematicians still have not been able to prove the conjecture. Mathematicians like Rihō Terras and Jeffery C. Lagarias have contributed heavily to the Conjecture, with Terras substantiating the idea of a coefficient stop time, which heavily simplifies the notion of a number's stop time, and Lagarias compiling what we know about the Conjecture with regards to the potential length of non-trivial Collatz Cycles (Lagarias, 1985). With the use of computers, we have proven that all numbers up to  $2^{68} = 2.95 * 10^{20}$  (295 quintillion) reach 1; however a proof remains highly elusive. Also, there is no mathematically describable pattern when considering total stopping time, the total number of steps before a number goes to one. Looking at Figure 1, while the total stopping time of a number  $T(n)$  generally appears to increase, we neither know if an upper limit exists, nor why some values of  $T(n)$  differentiate wildly from others. The conjecture's truth is predicated on the existence of a counterexample, which must become trapped in a non-trivial cycle. What would that cycle look like, if it even exists? What cannot be true about such a cycle? What does current mathematics say as of now? Using Terras' definition of the coefficient stop time, we look at the coefficient, i.e.  $3^O/2^E$ , ( $O$  = odd steps,  $E$  = even steps) and see after how many steps will it be less than 1. The coefficient stop time  $K(n)$  is defined as the first instance where  $3^O/2^E < 1$ . Rearranging the equation thus gives us the ratio of odd to even steps in such a theoretical cycle:  $\log_3 2 = 0.6309$ . Terras' proof demonstrates that for all  $n \geq 2$ , at most a finite number of  $n$  with  $K(n) \leq k$  for some constant  $k$  has a stopping time  $\sigma(n) \neq K(n)$ . Lynn E. Garner takes the concept of coefficient stopping time further; If a cycle exists, it has at least  $m$  odd terms such that  $m \geq \min(M, (n/2)3^{1-B(M)}(1 - 3^{-b(M)})^{-1})$ , for  $M \in \mathbb{N}$  such that  $b(M) = \max_{j \leq M} (-\log_3(1 - 2^{j \log_2 3} 3^{-j}))$  and  $B(M)_{j \leq M} = \max(-\log_3(2^{1+j \log_2 3} 3^{-j} - 1))$

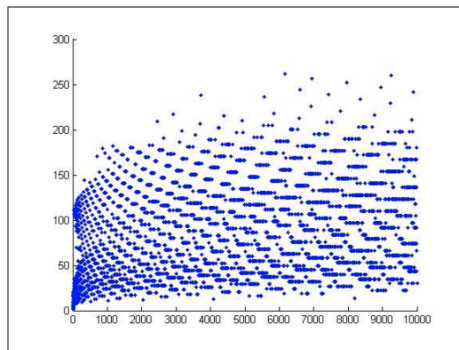


Figure 1: Total Stopping Time up to  $n=10000$

(Garner, 1981). With these functions, if we assume the least value of the cycle is  $n \geq 60,000,000$ , and using  $M = 14000$ , we get  $k \geq 35000$ . For  $n \geq 2,000,000,000$  and  $M = 41000$ , we get  $k \geq 105,000$ . Thus, a non-trivial cycle, if it exists, must have more than 105,000 odd terms. Such results have been improved upon with time, such that we would know the length of a theoretical non-trivial cycle. Eliahou defines  $Card \Omega$  as the size or length of a cycle  $\Omega$ ,  $\Omega_1$  as the set of odd terms, and  $m = \min \Omega$  as the minimum value in said cycle, such that  $\log_2 3 < Card \Omega / Card \Omega_1 < \log_2(3 + m^{-1})$  (Eliahou, 1993). Using  $m = 2^{40}$ , he proves that a nontrivial cycle must have more than 17087915 terms, giving one part of his equation,  $Card \Omega = 310994a + 17087915b + 85137581c$  for  $b > 0$ , and  $ac = 0$ , for  $a, b, c \in \mathbb{N}$ . 310994 comes from using  $m = 2^{39}$  as the minimum value in the cycle. The third term comes from the use of Farey pairs, but this formula is highly useful if to determine the length of a non-trivial cycle. To finish his proof, he calculates  $Card \Omega$  with  $m = 2^{1000}$ , giving  $1.05 * 2^{500}$ . While the scope of the Collatz Conjecture is outside of current mathematical methods, what we do understand about the Collatz Conjecture is less about infeasibly proving each number reaches 1 and more about proving (or disproving) the existence of cycles. If such a cycle exists, its coefficient stopping time would be astronomically large, and the further mathematicians investigate, the higher the lower bound for a non-trivial cycle's length will be.

## 2 References

Shalom Eliahou, The  $3x+1$  problem: new lower bounds on nontrivial cycle lengths, Discrete Mathematics, Volume 118, Issues 1–3, 1993, Pages 45–56, ISSN 0012-365X, [https://doi.org/10.1016/0012-365X\(93\)90052-U](https://doi.org/10.1016/0012-365X(93)90052-U). Garner, Lynn E., <https://www.ams.org/journals/proc/1981-082-01/S0002-9939-1981-0603593-2/S0002-9939-1981-0603593-2.pdf> Lagarias, Jeffrey C. “The  $3x + 1$  Problem and Its Generalizations.” *The American Mathematical Monthly*, vol. 92, no. 1, 1985, pp. 3–23. *JSTOR*, <https://doi.org/10.2307/2322189>. Accessed 2 May 2025.